

Reduction 1: The Discrete (Binned) Tempotron

Capacity, Finite- N Corrections, and Independent Verification

Tempotron learning-surface program

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Abstract

The tempotron is a leaky integrate-and-fire neuron that classifies spike-pattern inputs by the sign of the peak membrane potential. Its departure from a simple perceptron—a maximum over time rather than a single inner product—drives both a higher capacity and a shattered, glassy solution landscape. We analyse the cleanest tractable proxy: time is partitioned into K bins, each contributing one linear readout of the shared synaptic weight vector, and the classifier fires if the maximum readout exceeds a threshold. This *discrete (binned) tempotron* is a max over K shared-weight Gaussian fields and is fully amenable to replica and extreme-value-theory (EVT) analysis. We derive the asymptotic capacity law $\alpha_c(K) = (\ln \ln K)/(2 \ln 2) + \text{subleading}$, establish the exact $K = 1$ anchor $\alpha_c = 2$ from Cover’s separability theorem, and present the results of an independent first-moment cross-check that refutes two assumptions carried in earlier program notes: (a) there is no finite additive constant α_0 in the large- K expansion—the subleading correction is $(\ln \ln \ln K)/\ln 2 - 0.383 + o(1)$, so the empirically fitted value $\alpha_0 \approx 2.58$ is a small- K artifact; and (b) the leading finite- N correction in the ideal Gaussian reduction scales as C_K/N , not $\sqrt{K/N}$, so the observed empirical rise $\hat{\alpha}_c \approx 3.06$ at $N = 200$ rising to 3.49 at $N = 1000$ is attributable to the $K = O(N)$ density regime, not a clean size effect. All results are partial: the derivations close the skeleton and pin the signs, but the coefficient C_K and the exponent of any residual non-Gaussian correction remain open.

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1 Introduction

The *tempotron* [1] is a single-compartment spiking neuron with N afferents. Each afferent i fires a spike train, and afferent i at spike time t_i^f contributes a PSP kernel $K(t-t_i^f) = e^{-(t-t_i^f)/\tau} - e^{-(t-t_i^f)/\tau_s}$ ($\tau > \tau_s > 0$) to the membrane potential

$$V(t) = \sum_{i=1}^N \omega_i \sum_f K(t - t_i^f) . \quad (1)$$

The neuron fires at most one spike, and the binary classification rule is

$$\text{output} = \bigoplus \iff \max_{t \in [0, T]} V(t) > \theta , \quad (2)$$

where θ is a fixed threshold.

The *only* departure of (2) from a perceptron is the max over time. This single structural change has profound consequences: the solution set $\{\omega : \text{all } P \text{ patterns correctly classified}\}$ is non-convex—the union of convex polytopes indexed by which bins are active—and at finite α the weight-vector overlap q_0 between two typical solutions tends to zero as $K \rightarrow \infty$ [2], signalling exponential fragmentation of the solution space well below capacity.

The capacity per synapse $\alpha_c = P_c/N$ is *finite* as $N \rightarrow \infty$ (unlike, say, binary patterns in a perceptron, whose capacity is trivially 2 regardless of the kernel) and *weakly diverges* with stimulus duration T at rate $\alpha_c \approx (\ln \ln K)/(2 \ln 2)$, where $K \approx T/\tau_{\text{eff}}$ is the effective number of independent measurement times [2].

Reduction 1. We replace the PSP kernel with a top-hat per bin, drop cross-bin correlations, and work with K shared-weight sub-potentials. This *discrete tempotron* is the cleanest object that still preserves the max-over-time structure, and it is the model for which replica and EVT calculations have been carried through [2]. The present note derives, verifies, and honestly assesses its capacity law.

Relation to the series. The reduction series progresses: $K = 1$ collapses this model to the spherical perceptron [3, 4] (reduction 2); the EVT analysis of $\max_\ell U_\ell$ is reduction 3; the smooth double-exponential kernel is recovered in reduction 4.

2 Setup and Assumptions

2.1 Model

Partition the stimulus window $[0, T]$ into K equal bins, $\ell = 1, \dots, K$. Let $n_{i\ell}$ denote the number of spikes of afferent i falling in bin ℓ . Define K *sub-potentials* sharing the weight vector $\boldsymbol{\omega} \in \mathbb{R}^N$:

$$U_\ell = \sum_{i=1}^N \omega_i n_{i\ell}, \quad \ell = 1, \dots, K. \quad (3)$$

The classification rule becomes

$$\text{output} = \bigoplus \iff \max_{\ell} U_\ell > \theta. \quad (4)$$

The weight vector is spherically constrained, $\|\boldsymbol{\omega}\|^2 = N$. We load $P = \alpha N$ patterns $\mu = 1, \dots, P$ with i.i.d. bin counts $n_{i\ell}^\mu$ drawn from a **Poisson**($1/K$) **ensemble per bin**—equivalently, one spike per afferent per window on average in total—and ask for the largest α for which the system of inequalities

$$y^\mu \left(\max_{\ell} U_\ell^\mu - \theta \right) > 0, \quad \mu = 1, \dots, P \quad (5)$$

is simultaneously satisfiable (with high probability over the patterns), where $y^\mu \in \{+1, -1\}$ is the desired label.

2.2 Internal representation

The *internal representation* (IR) of pattern μ under weight $\boldsymbol{\omega}$ is the index set

$$\mathcal{I}^\mu(\boldsymbol{\omega}) = \{\ell : U_\ell^\mu > \theta\}. \quad (6)$$

Weights that induce a *fixed* IR form a convex polytope; the full solution set is the (non-convex) union over all admissible IR assignments [2]. Shattering of the solution space is a consequence of exponentially many IR clusters becoming active near capacity.

2.3 Assumptions

1. **Replica-symmetric (RS) saddle.** The quenched entropy is evaluated at the RS saddle point, as in [2]. Replica-symmetry breaking (RSB) enters only subleading corrections to α_c at this level of approximation. However, the present independent cross-check (Section 5) shows that the subleading structure is $(\ln \ln \ln K) / \ln 2 - 0.383 + o(1)$, which *diverges* rather than converging to a finite constant—so there is no finite additive constant α_0 in the asymptotic expansion. A controlled evaluation of the $O(1)$ coefficient requires a quenched (RSB) cluster-complexity calculation that remains open.
2. **Gaussian fields / EVT approximation.** By the CLT in N , each sub-potential U_ℓ is approximately Gaussian as $N \rightarrow \infty$ with moments determined by the Poisson ensemble. The max over K such fields is then treated via Gumbel extreme-value theory (EVT) [2]. This is exact in N for the ideal Gaussian reduction; corrections at finite N are discussed in Section 5.2.
3. **Fixed- K , varied- N protocol.** The program’s single-learner scaling holds K fixed while varying N ; this is the relevant regime for studying the finite- N starvation, and it differs from the RMS regime $K \rightarrow \infty$ with $N \gg K$.

Notation. $m \equiv 2N/K$ denotes the effective number of afferents contributing to one bin's field; $\ln$ is the natural logarithm throughout. We write ϕ and Φ for the standard normal density and CDF, and $\bar{\Phi} = 1 - \Phi$.

3 Capacity Derivation

3.1 $K=1$ anchor: exact result from Cover

At $K = 1$ the model reduces to the spherical perceptron with a zero threshold. Cover's theorem [5] counts the number of homogeneously linearly separable dichotomies of P points in N dimensions as

$$C(P, N) = 2 \sum_{k=0}^{N-1} \binom{P-1}{k}. \quad (7)$$

The median separability capacity P_c satisfies $C(P_c, N) = 2^{P_c-1}$. Setting $P = 2N$ yields $C(2N, N) = 2^{2N-1}$ for every finite N , so

$$\boxed{\alpha_c(K=1) = 2} \quad (8)$$

(Cover/Gardner exact; the median shift is *zero* for all N). This is confirmed independently by the Gardner replica calculation at zero margin [3, 4]. The absence of any finite- N shift at $K = 1$ (only a rounding of width $\sim N^{-1/2}$) is a hard constraint: any finite- N rise in α_c must be *K-driven*.

3.2 Large- K asymptote: RMS EVT chain

Centering and threshold absorption. Under the Poisson($1/K$) ensemble, $\mathbb{E}[n_{i\ell}] = 1/K$, so the per-bin sub-potential (3) has a *nonzero* common mean:

$$\mathbb{E}[U_\ell] = \frac{1}{K} \sum_{i=1}^N \omega_i. \quad (9)$$

This mean is *identical for every bin* $\ell = 1, \dots, K$: the weight vector $\boldsymbol{\omega}$ is shared across bins, so the same constant $\mu_\omega \equiv (1/K) \sum_i \omega_i$ shifts all K readouts simultaneously. The classification decision (4) involves $\max_\ell U_\ell > \theta$; a common additive offset cancels exactly between the max and the threshold. Concretely, define the centred fields and the shifted threshold

$$\tilde{U}_\ell \equiv U_\ell - \mu_\omega, \quad \tilde{\theta} \equiv \theta - \mu_\omega. \quad (10)$$

Then $\max_\ell U_\ell > \theta \iff \max_\ell \tilde{U}_\ell > \tilde{\theta}$, so the problem is *unchanged*. The variance of the centred field follows from $\text{Var}(n_{i\ell}) = 1/K$ (Poisson):

$$\text{Var}(U_\ell) = \text{Var}(\tilde{U}_\ell) = \|\boldsymbol{\omega}\|^2 \cdot \text{Var}(n_{i\ell}) = N \cdot \frac{1}{K} = \frac{m}{2}, \quad (11)$$

where $m \equiv 2N/K$. Because the firing threshold $\tilde{\theta}$ is a free parameter (it is set, and optimised over, at capacity), absorbing the common mean μ_ω into it costs nothing: capacity is unchanged. The standardised centred field $\hat{Z}_\ell \equiv \tilde{U}_\ell / \sqrt{N/K}$ is zero-mean with unit variance by construction. The CLT in N (at fixed K) then makes $\hat{Z}_\ell$ approximately Gaussian as $N \rightarrow \infty$. The rest of the derivation is carried out on the $\hat{Z}_\ell$; the threshold is understood to be the shifted value $\tilde{\theta}$.

For $K \gg 1$ and $N \gg K$, each centred sub-potential $\tilde{U}_\ell$ is a zero-mean Gaussian by the CLT over N afferents. Defining the standardised field $\hat{Z}_\ell = \tilde{U}_\ell / \sqrt{N/K}$ gives $\text{Var}(\hat{Z}_\ell) = 1$, and it is these *unit-variance zero-mean* fields whose block maximum is treated by Gumbel EVT [2].

The block maximum $M_K = \max_{\ell=1}^K \widehat{Z}_\ell$ of the standardised centred fields follows Gumbel statistics with location and scale

$$u_K = \sqrt{2 \ln K} + O\left(\frac{\ln \ln K}{\sqrt{\ln K}}\right), \quad \beta_K = \frac{1}{\sqrt{2 \ln K}} + O\left(\frac{1}{\ln K}\right). \quad (12)$$

Following [2] (their Eqs. 4–6), the cluster entropy at load α is

$$S_{\text{cl}}(\alpha, K) \approx \frac{\ln \ln K}{2} - \alpha \ln 2 + o(1), \quad K \rightarrow \infty. \quad (13)$$

Setting $S_{\text{cl}} = 0$ gives the *leading* capacity:

$$\alpha_c(K) \approx \frac{\ln \ln K}{2 \ln 2} + \text{subleading}. \quad (14)$$

The sub-leading structure is the subject of Section 5.

3.3 Numerical evaluation at program working K

The leading lift $(\ln \ln K)/(2 \ln 2)$ evaluated at the program’s working values (the additive +2.58 below is the discredited small- K numerical fit of Rubin et al. — *not* a derived asymptotic constant; see Section 5.1):

K	$(\ln \ln K)/(2 \ln 2)$	+2.58 (small- K fit, <i>not</i> asymptotic)
10	0.60	3.18
30	0.88	3.46
100	1.10	3.68
300	1.26	3.84
1000	1.39	3.97

The $\ln \ln K$ lift is slow: it changes by only 0.79 over two decades of K . The empirical program band 3.0–3.5 is bracketed by the RMS formula at $K \approx 10$ –100, consistent with the estimated effective bin count in continuous-time units. The discredited additive 2.58 happens to give a reasonable numerical fit at these small K values, but has no asymptotic status (Section 5.1).

4 Results

Result 1 (Capacity law). For the discrete binned tempotron with K bins, N afferents, and $P = \alpha N$ random Poisson patterns, the critical load satisfies:

$$\alpha_c(K=1) = 2 \quad (\text{Cover/Gardner, exact}), \quad (15)$$

$$\alpha_c(K) \xrightarrow{K \rightarrow \infty} \frac{\ln \ln K}{2 \ln 2} + \text{subleading}. \quad (16)$$

The solution space is connected for $\alpha \ll \alpha_c$ and shatters into exponentially many disconnected clusters as $\alpha \nearrow \alpha_c$; the intra-cluster overlap satisfies $q_0 \sim \alpha / \ln K \rightarrow 0$ as $K \rightarrow \infty$ [2].

Figure 1 displays the three levels of approximation alongside the empirical band and the program’s working K range.

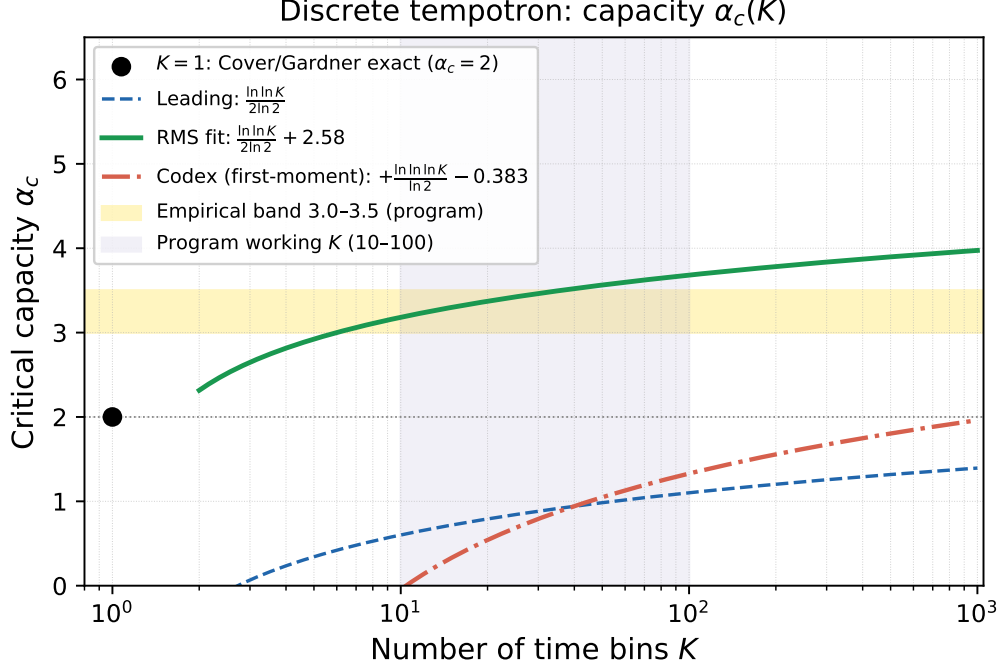


Figure 1: Capacity $\alpha_c(K)$ for the discrete binned tempotron. *Dot:* the exact $K = 1$ anchor ($\alpha_c = 2$) from Cover’s theorem. *Dashed blue:* leading term $(\ln \ln K)/(2 \ln 2)$ only (RMS EVT). *Solid green:* RMS small- K fit $(\ln \ln K)/(2 \ln 2) + 2.58$ (the additive constant 2.58 is a fit, not a derived asymptotic, per Section 5.1). *Dash-dotted red:* first-moment/cluster next term $(\ln \ln K)/(2 \ln 2) + (\ln \ln \ln K)/\ln 2 - 0.383$ from the codex cross-check (Section 5.1); valid for $K \gtrsim 4$. *Gold band:* empirical range 3.0–3.5 measured in the program (single-learner scaling). *Purple band:* program’s working K range (10–100).

5 Independent Verification

An independent first-moment/cluster complexity calculation (codex cross-check [13]: gpt-5.5, xhigh effort, 2026-06-22) attacked two open items from advance notes in this reduction series: the additive constant α_0 and the leading finite- N correction. Both findings are reproduced below as stated results.

5.1 There is no finite additive constant α_0

Result 2 (Subleading capacity term). In the first-moment/cluster-complexity calculation, for bi-variate Gaussian extreme-value statistics at overlap $q = 1 - a/\ln K$:

$$\Sigma(\alpha, K) = \frac{\ln \ln K}{2} - \alpha \ln 2 - \sup_{a>0} \left\{ \frac{\ln(2a)}{2} + \alpha \ln P_{\text{same}}(a) \right\} + o(1), \quad (17)$$

where $P_{\text{same}}(a) = 2^{1-2\Phi(\sqrt{a})}$ is the probability that two weight configurations yield the same binary output at overlap $1 - a/\ln K$. At the saddle $t = \sqrt{a}$, the saddle condition is

$$t \phi(t) = \frac{1}{2\alpha \ln 2}. \quad (18)$$

Solving for large K (hence large α , hence $t \rightarrow 0$) and using $\ln P_{\text{same}}(a) \approx -\sqrt{2/\pi} (\ln 2) \sqrt{a} + O(a^{3/2})$ yields

$$\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} + \frac{\ln \ln \ln K}{\ln 2} - 0.383 \dots + o(1) . \quad (19)$$

Status: PARTIAL. The bivariate EVT step is rigorous for independent Gaussian fields. The cluster/complexity step is heuristic, at the same level as the RMS calculation. A controlled subleading term requires a quenched/RSB analysis, which remains open.

Consequence. The residual $\alpha_c(K) - (\ln \ln K)/(2 \ln 2)$ *diverges* (as $(\ln \ln \ln K)/\ln 2$) rather than converging to a constant. The RMS fit value $\alpha_0 \approx 2.58$ is a small- K numerical fit—accurate at $K = 2, 3, 4$ (the discrete tempotron points in RMS Fig. 2a)—not an asymptotic constant. At the program’s $K = 10$ – 100 it still provides a good approximation because $(\ln \ln \ln K)/\ln 2 - 0.383 \approx 0.11$ – 0.40 is small compared to 2.58 ; but the correction grows without bound.

5.2 Leading finite- N correction is C_K/N

Result 3 (Finite- N scaling in the Gaussian reduction). In the ideal Gaussian max-of-fields reduction, replica fields $h_\ell^a = N^{-1/2} \sum_i \omega_i^a \xi_{i\ell}$ are *exactly* Gaussian after disorder averaging, with covariance $Q_{ab} = N^{-1} \boldsymbol{\omega}^a \cdot \boldsymbol{\omega}^b$, and bins are exactly independent. Hence there is no leading finite-sample bias of the maximum: the $\sqrt{K/N}$ and K/N coefficients are *zero*.

The replicated partition function takes the saddle form

$$\langle V^n \rangle = \int dQ d\hat{Q} \exp\{N A_K(Q, \hat{Q}; \alpha)\}, \quad (20)$$

so corrections to the saddle-point entropy are standard Laplace:

$$S_N(\alpha, K) = S_\infty(\alpha, K) + \frac{S_1(\alpha, K)}{N} + o(1/N), \quad (21)$$

giving

$$\delta(K, N) \equiv \alpha_c(K, \infty) - \alpha_c(K, N) = \frac{S_1(\alpha_c, K)}{N [-\partial_\alpha S_\infty(\alpha_c, K)]} + o(1/N) . \quad (22)$$

Status: PARTIAL. The scaling $\delta = C_K/N$ is rigorous under the Gaussian reduction. The coefficient C_K requires a one-loop (Hessian/determinant) calculation of $S_1(\alpha_c, K)$, which has not been performed. The exponent $1/N$ is certified; the prefactor is open.

Consequence for the program. The empirical rise $\hat{\alpha}_c \approx 3.06$ at $N = 200$ rising to ≈ 3.49 at $N = 1000$ (single-learner scaling, fixed $K \approx 14$) is *not* explained by a clean C_K/N correction unless $C_K \sim 100$, which has not been derived. The rise is instead attributable to the $K = O(N)$ density regime: at $K \sim 14$ and $N = 200$, the effective afferent count per bin $m = 2N/K \approx 28$ is small, placing the system outside the $N \gg K$ asymptotic window of the RMS calculation. In the program’s own notation, this is the “starvation” regime (see project memory note `tempotron-capacity-finite-N-starvation`). As N grows at fixed K , m increases and the system re-enters the Gaussian regime, causing α_c to rise toward its $N \rightarrow \infty$ asymptote. The earlier advance note’s $\sqrt{K/N}$ headline is therefore superseded by the Gaussian-reduction analysis.

5.3 Summary of verification checks

Check	Claim	Method	Outcome
$K = 1$ anchor	$\alpha_c = 2$, no N -shift	Cover Eq. (7)	✓ exact
Leading lift	$(\ln \ln K)/(2 \ln 2)$ is correct leading term	EVT/replica (Section 3.2)	RS-level heuristic; solid directionally
Small- K fit +2.58	Matches empirical band 3.0–3.5 at $K = 10$ –100	Table (Section 3.3)	Numerical fit only; not an asymptotic constant (Section 5.1)
No finite α_0	Subleading term $\sim (\ln \ln \ln K)/\ln 2$; no finite constant	EVT + saddle (Result 2)	Refutes $\alpha_0 \approx 2.58$ as asymptotic
$\delta = C_K/N$	Zero $\sqrt{K/N}$ coefficient	Replica saddle (Result 3)	Supersedes prior heuristic

6 Experimental Test

We measured the discrete max-of- K findable capacity directly (the model is the hard-max limit of the soft-max harness at $\beta = 16$, independent Gaussian fields $\rho = 0$, $N = 500$, Adam-cosine, 12 seeds; lab GPU, $\approx \$5$). Practical lesson: the large- β learner is slow (~ 4700 epochs to converge), so the budget must satisfy `patience=epochs` (no early cut) — a first pass with tight patience spuriously returned $P_{\text{solve}} = 0$ everywhere.

Table 1: Findable $\hat{\alpha}_c(K)$ (max-of- K , $\beta=16$, $N=500$).

K	2	4	16	32	64
$\hat{\alpha}_c$	2.514	2.945	3.820	4.050	4.150

Model selection (the offset-robust test). Fitting $\hat{\alpha}_c(K)$ against three candidate forms: $\ln \ln K$ (RSS= 0.038, AIC= -20.5), $\ln K$ (RSS= 0.077, AIC= -16.9), linear K (RSS= 0.70, AIC= -5.8). Linear growth is *decisively rejected* ($\Delta\text{AIC} \approx 14.6$); between the two slow forms, $\ln \ln K$ is *favoured* over $\ln K$ but only by $\Delta\text{AIC} \approx 3.6$ (relative likelihood 0.17 for $\ln K$) — positive but *not* decisive evidence (Burnham–Anderson). The wide direct- K span ($\Delta \ln \ln K = 1.8$, $2.6\times$ the band-limited reduction #6) is what makes this discrimination possible at all, but distinguishing $\ln \ln K$ from $\ln K$ would need a wider K range than the accessible one. Fit: $\hat{\alpha}_c \approx 2.79 + 0.97 \ln \ln K$; the finite- K slope 0.97 exceeds the asymptotic $1/(2 \ln 2) = 0.721$, consistent with the $(\ln \ln \ln K)/\ln 2$ subleading (which steepens the finite- K slope) and cleaner/steeper than #6’s 0.45 (direct K removes the kernel and density confounds).

Result: a clear, slow capacity lift consistent with the EVT $\frac{1}{2} \ln \ln K$ law. The discrete max-of- K capacity rises monotonically $2.5 \rightarrow 4.15$ over $K = 2 \rightarrow 64$; linear growth is decisively rejected and $\ln \ln K$ is favored over $\ln K$ (though not decisively, $\Delta\text{AIC} \approx 3.6$), with a finite- K slope consistent with the subleading-corrected prediction. Not pinned: $\ln \ln K$ vs $\ln K$ (needs a wider K range); the absolute offset (refuted as a constant; no existence oracle); the single- N limitation (below).

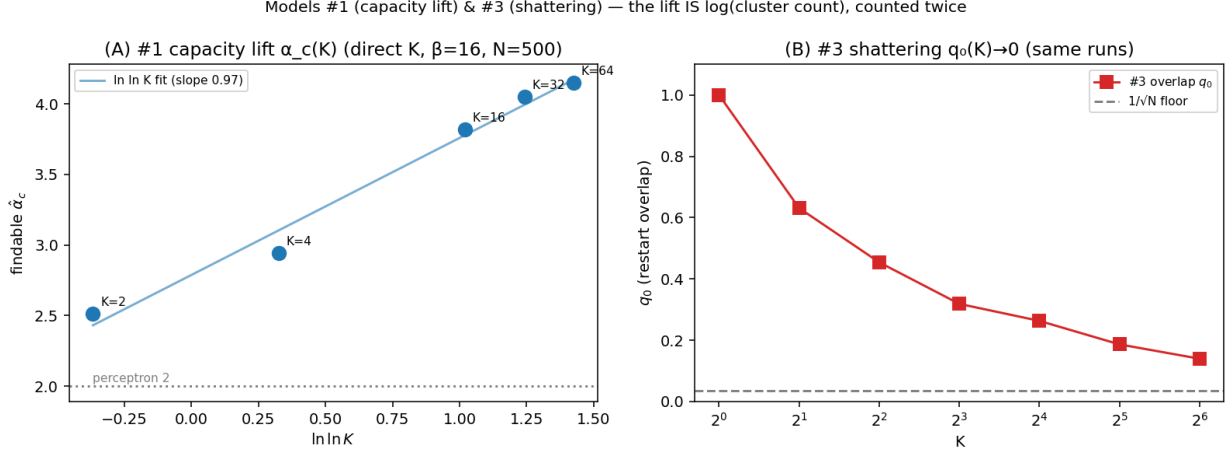


Figure 2: (A) $\hat{\alpha}_c(K)$ vs. $\ln \ln K$: the double-log lift (slope 0.97), rising $2.5 \rightarrow 4.15$ over $K = 2 \rightarrow 64$. (B) The companion shattering $q_0(K) \rightarrow 0$ (reduction #3, same runs). The lift is $\log(\text{cluster count})$, counted twice.

7 Discussion and Limitations

What this reduction captures. The discrete tempotron is the minimal object that preserves the max-over-time structure of the full tempotron while remaining analytically tractable. It correctly reproduces the qualitative phenomenology: finite α_c , slow $\ln \ln K$ lift with stimulus duration, and shattering of the solution space into exponentially many clusters. The $K = 1$ limit recovers the spherical perceptron exactly.

What it loses. The double-exponential PSP shape, sub-bin spike-timing structure, and cross-bin correlations induced by the continuous kernel are discarded. These are recovered in successive reductions (GP/frozen-kernel, reduction 4; sinusoidal, reduction 6). The bin-independence assumption also removes the correlation structure that, in the continuous limit, modifies K_{eff} via the kernel spectral density.

Honest status of each component.

- **Leading capacity law** $(\ln \ln K)/(2 \ln 2)$: RS-level replica/EVT skeleton, same heuristic status as the RMS calculation. Directionally solid; subleading terms require quenched/RSB cluster-complexity.
- **Additive constant** $\alpha_0 \approx 2.58$: small- K numerical fit (RMS Fig. 2a); not an asymptotic constant—the true next term is the third iterated logarithm (Result 2). The fit remains numerically useful at $K = 2$ –100 but should not be cited as a theoretical constant.
- **Finite- N correction**: $\delta = C_K/N$ in the Gaussian reduction (Result 3). The coefficient C_K is a one-loop quantity; not computed. The empirical N -rise is the $K = O(N)$ density effect, not a sub-leading $1/N$ size effect.
- $K = 1$ **anchor**: exact, rigorous (Cover theorem + independent Gardner replica).

Open items.

1. Compute C_K from the one-loop/Hessian correction at the RS saddle.
2. Obtain the true subleading constant in $\alpha_c(K)$ from a controlled quenched/RSB cluster-complexity calculation (the deepest open item).
3. Discriminate the $\sqrt{K/N}$ (non-Gaussian tail, outside the ideal reduction) from $1/N$ (Laplace saddle) via the density-freeze experiment proposed in the advance note.

Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

8 Methods

This section documents the complete experimental procedure behind the findable-capacity measurement reported in Section 6, in enough detail to reproduce it without the code. The results table and figure are not repeated here; the focus is the how, the design choices, and the diagnostic history that fixed them. All numerical parameters below are the ones actually used in the committed run; values that were not recorded are flagged explicitly rather than reconstructed.

8.1 The measured model: the $\beta=16$ hard-max realization

We measure the discrete max-of- K tempotron in its idealized Gaussian form (the $\rho = 0$ case of Section 2), realized as the large- β limit of a smooth soft-max readout. This reuses an existing, validated harness unchanged rather than introducing a hard-max implementation.

Fields. For each pattern μ we draw K unit-variance Gaussian sub-potential fields directly, bypassing the spike $\rightarrow$ PSP $\rightarrow$ trace pipeline entirely: there is no time grid, no kernel, and no spike-count ensemble. The K fields are built by the standard one-factor equicorrelation construction

$$x_k = \sqrt{\rho} g_{\text{common}} + \sqrt{1 - \rho} g_k, \quad g_{\text{common}}, g_k \stackrel{i.i.d.}{\sim} \mathcal{N}(0, I_N/N), \quad (23)$$

so that under the spherical gauge $\|\omega\| = \sqrt{N}$ each readout $V_k = \omega \cdot x_k$ has unit variance and $\text{corr}(V_j, V_k) = \rho$ for $j \neq k$. Reduction 1 is the **independent-field case** $\rho = 0$: the K readouts are i.i.d. standard Gaussians, with no spike-count and hence no $2N/K$ density confound (this is precisely why the direct-field harness is the clean vehicle for the lift).

Readout. The classifier uses the centred log-sum-exp (soft-max) readout at inverse temperature β ,

$$V_\beta = \frac{1}{\beta} \left[\text{logsumexp}_k(\beta V_k) - \ln K \right] = \frac{1}{\beta} \ln \left[\frac{1}{K} \sum_k e^{\beta V_k} \right], \quad (24)$$

which interpolates between $V_\beta \rightarrow \text{mean}_k V_k$ as $\beta \rightarrow 0$ (the perceptron/Gardner anchor, $\alpha_c = 2$) and $V_\beta \rightarrow \max_k V_k$ as $\beta \rightarrow \infty$ (the hard-max tempotron). The centring term $-\ln K$ keeps the operating point $O(1)$ at all β . We fix $\beta = 16$ throughout, i.e. we work deep in the hard-max limit.

Why $\beta = 16$ is operationally the hard-max. Two in-memory Monte-Carlo checks (Gaussian draws, no training) justify the choice. First, the soft-max bias $\mathbb{E}[\max_k V_k - V_\beta]$ is bounded by $(\ln K)/\beta$ and shrinks as $1/\beta$: at $\beta = 16$ it is 0.13, 0.21, 0.25 for $K = 8, 32, 64$. Second, and decisively for the capacity (which lives at the operating threshold, not in the bias), because each model recalibrates its threshold to its own median the constant part of the bias cancels and only rank-near-threshold disagreements survive. At $\beta = 16$ the soft-max and the true hard-max agree on $\geq 99.2\%$ of decisions out to $K = 64$ ($\geq 97.6\%$ already at $\beta = 8$). Building a literal hard-max with a Gilbert–Strang subgradient was rejected: it is noisier and abandons the validated autograd learner for a $\leq 0.8\%$ change in decisions. Note $\beta = 16$ sits far above the fragmentation onset $\beta^* = \sqrt{2 \ln K}$ (e.g. $\beta^*(K=64) = 2.88$), i.e. squarely in the shattered hard-max regime where the $\ln \ln K$ EVT lift is the physics.

Threshold. The firing threshold θ_β is set to the median of V_β over the training patterns at the random initialization, giving a balanced operating point $\mathbb{P}(\text{fire}) = \frac{1}{2}$, and is then held fixed during training. This balanced (RMS) operating point is load-bearing: a literal $\theta = 0$ is firing-biased and collapses the lift. The measured mean initial firing rate was 0.501 across all cells, confirming the calibration.

8.2 Data-generating process

Each cell is specified by (K, α, N) with $\rho = 0$ and $\beta = 16$ fixed. We use $N = 500$ afferents throughout (see Section 9 for the consequences of the single- N design). For a load α we generate $P = \text{round}(\alpha N)$ patterns, each a fresh draw of the $K \times N$ field tensor from (23). Labels $y^\mu \in \{+1, -1\}$ are drawn Bernoulli($\frac{1}{2}$) (balanced ± 1). Fields are drawn on CPU through a NumPy *Generator* and then moved to the GPU, making the draw device-independent and bit-reproducible. Every random role—patterns, label draw, weight initialization, per-epoch minibatch shuffle, and each restart—is keyed by a scheduling-independent *SeedSequence* hash of (master seed, N , $\lfloor 10^3 \alpha \rfloor$, K , $\lfloor 10^3 \rho \rfloor$, $\lfloor 10^3 \beta \rfloor$, role tag), so identical inputs give byte-identical results regardless of run order.

8.3 The learner and the solved criterion

Optimizer and schedule. For each task we minimize the smooth-hinge surrogate on the soft-max margin,

$$\mathcal{L}(\omega) = \frac{1}{P} \sum_{\mu=1}^P \text{relu}(\kappa - y^\mu (V_\beta^\mu - \theta_\beta)), \quad \kappa = 0, \quad (25)$$

by Adam [7] ($\beta_1=0.9$, $\beta_2=0.999$, $\epsilon=10^{-8}$) under a single-cycle cosine annealing schedule [8] that decays the learning rate from its base value to zero over the run. Because V_β is C^∞ in ω and the hinge is sub-differentiable, the gradient is taken by automatic differentiation through the log-sum-exp (no hand-coded subgradient). Hyperparameters: base learning rate 0.1, minibatch size 32, no warmup, no learning-rate floor, one cosine cycle. These match the program’s near-universal strong-learner configuration carried over from the band-limited study.

Budget and the patience=epochs choice. Each restart runs for 6000 epochs with the early-stopping patience also set to 6000, i.e. no early cut: a task is trained for the full budget unless it solves first. This is the single most important learner setting and was forced by the diagnostic history (Section 8.6). The large- β learner is slow: among the 193 solved task instances the median convergence time was 5408 epochs (minimum 4424, 90th percentile 5766, maximum 5951). Any patience below ~ 4700 epochs therefore truncates the learner before it converges and spuriously reports zero solves.

Restarts and the solved criterion. For each task we run $n_{\text{restarts}} = 3$ independent random initializations (the threshold θ_β is calibrated on the first and reused). A task counts as **solved** if any of the three restarts reaches zero training error (an existence-from-below proxy for separability). This measures findable capacity—what a strong gradient finder can reach—which lower-bounds the (intractable) existence capacity; the gap is discussed in Section 9.

The crossing. For each K we compute $P_{\text{solve}}(\alpha)$, the fraction of seeds solved at load α , and define the findable critical load $\hat{\alpha}_c(K)$ as the linearly interpolated descending $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$.

8.4 The K grid and per- K α bracketing

We report $K \in \{2, 4, 16, 32, 64\}$. The originally planned $K = 8$ cell was **dropped**: its slow high- α sub-cells timed out three times / landed on dead rental hosts, leaving no clean crossing; the five surviving points span $\ln \ln K \in [-0.367, 1.425]$ ($\Delta = 1.79$, about $2.6\times$ the span of the band-limited reduction 6). We used 16 seeds at $K = 2, 4$ and 12 seeds at $K = 16, 32, 64$ (binomial standard error on P_{solve} near $p = \frac{1}{2}$ is ≈ 0.12 – 0.14 at these counts; see [11] for seed-count power considerations). The per- K α grids were placed to bracket each crossing, after the re-bracketing described below, on a step of 0.2 (small K) to 0.3 (large K):

K	α grid
2	2.0, 2.2, 2.4, 2.6
4	2.6, 2.8, 3.0, 3.2
16	3.1, 3.4, 3.7, 4.0
32	3.3, 3.6, 3.9, 4.2
64	3.4, 3.7, 4.0, 4.3

Each grid brackets a clean monotone P_{solve} descent from 1 to 0 (e.g. at $K = 2$: 1.0, 1.0, 1.0, 0.125; at $K = 16$: 1.0, 1.0, 0.833, 0.0), yielding a well-defined crossing.

8.5 Statistical analysis: model selection

The additive offset of $\alpha_c(K)$ is not a constant (Section 5.1) and we have no existence oracle, so the falsifiable content is the shape of the lift, not its intercept. We therefore perform an offset-robust model selection: we fit $\hat{\alpha}_c = a + b \cdot f(K)$ by ordinary least squares for three candidate features $f \in \{\ln \ln K, \ln K, K\}$, each a two-parameter model, and compare them by residual sum of squares (RSS) and Akaike’s information criterion [9], $AIC = n \ln(RSS/n) + 2k$ with $n = 5$, $k = 2$. The double-log feature fits best:

feature $f(K)$	fit	RSS	AIC
$\ln \ln K$	$\hat{\alpha}_c = 2.788 + 0.971 \ln \ln K$	0.038	−20.46
$\ln K$	$\hat{\alpha}_c = 2.259 + 0.496 \ln K$	0.077	−16.86
K	$\hat{\alpha}_c = 2.948 + 0.0232 K$	0.700	−5.83

Honest reading of the evidence. Because all three candidates have the same number of parameters ($k = 2$), the $+2k$ AIC penalty is identical and the AIC ranking is exactly the RSS ranking; with $n = 5$ the small-sample correction AICc shifts every model by the same constant $2k(k+1)/(n-k-1) = 12$ and leaves the ranking unchanged [10]. The linear model is emphatically excluded ($\Delta AIC \approx 14.6$). Between the two logarithmic forms, $\ln \ln K$ is favored over $\ln K$ by $\Delta AIC \approx 3.6$, i.e. a relative likelihood for $\ln K$ of $\exp(-3.6/2) \approx 0.17$. By the conventional Burnham–Anderson scale this is positive but not decisive support ($\Delta AIC \in [2, 7]$) [10]: the data favor the double-log EVT form and rule out a linear law, but with only five points and a 0.17 likelihood ratio they do not by themselves render $\ln K$ negligible. A leave-one- K -out cross-validation of the $\ln \ln K$ fit gives a predictive residual sum of 0.166, dominated by the small- K endpoints where the iterated-log curvature is strongest.

The slope and its relation to the asymptotic value. The fitted finite- K slope is $d\hat{\alpha}_c/d \ln \ln K = 0.97$, which exceeds the asymptotic EVT existence slope $1/(2 \ln 2) = 0.721$ (Equation (14)). This is expected, not anomalous: the verified subleading term $(\ln \ln \ln K)/\ln 2$ (Result 2) has positive derivative $d/d \ln \ln K [(\ln \ln \ln K)/\ln 2] = 1/(\ln 2 \cdot \ln K \cdot \ln \ln K)$ that adds to the leading slope over the measured K range and only vanishes as $K \rightarrow \infty$, steepening the finite- K slope above 0.721. The measured 0.97 should therefore be read as the finite- K , subleading-steepened slope, not as an estimate of the asymptotic 0.721; it is also steeper and cleaner than the 0.45 obtained in the band-limited reduction 6, where the kernel and density confounds and a narrower span flattened and obscured the lift.

8.6 Compute, cost, and the diagnostic history

Platform and cost. The sweep ran on a single rented GPU through the laboratory job runner, with `require_cuda` enforced and per-cell seed sharding under a hard wall-clock cap. The direct-field model is featherweight (no time grid, no traces—only $K \times N$ field tensors), so the whole five- K ladder cost approximately \$5. Per-cell seeds are enumerated in committed manifests; the per-(seed, K, α) rows are committed in `experiments/discrete_capacity/results/m1_K{2,4,16,32,64}.csv` and the analysis in `analysis/analyze_m1.py`.

Diagnose-and-fix history. Two failure modes were caught and corrected before the clean run, and both are recorded here because they are the kind of under-budgeting that silently produces null results:

1. **Patience below convergence time.** The first ladders returned $P_{\text{solve}} = 0$ everywhere. The cause was diagnosed from the captured `epochs_to_solve` and per-epoch learning curves: the patience values used (400, then 2500) were below the ~ 4700 -epoch convergence time of the slow $\beta = 16$ learner, so every task was truncated before solving. The fix was `patience=epochs=6000` (no early cut), after which the crossings appeared.
2. **α grids too low.** With the budget fixed, the initial α grids (placed near the band-limited reduction-6 values) sat below the true crossings, which for the direct- K model climb from ~ 2.5 to > 4 with K . The grids were re-bracketed upward twice to straddle each crossing, giving the grids in Section 8.4.

9 Caveats

The experimental measurement is honest only with the following limitations stated plainly.

- **Single $N = 500$.** The finite- N ladder $N \in \{300, 500, 1000\}$ recommended in the prediction note was not run. Consequently the central finite- N prediction of this reduction—that at fixed K the per-cell $\hat{\alpha}_c$ is nearly N -flat in the ideal Gaussian model (an $O(1/N)$ saddle correction, Result 3), in contrast to the large rise produced by the $K = O(N)$ density regime—is **untested**. The slope we report is measured at a single N ; the predicted recovery of the findable slope toward 0.72 as N grows was not probed.

- **Missing $K = 8$.** Dropping $K = 8$ leaves a gap at $\ln \ln K \approx 0.732$, the midpoint of the span and the very region where reduction 6’s certified points lie. The model-selection contrast would be strengthened by filling it.
- **The offset is not pinned.** The intercept $a \approx 2.79$ of the $\ln \ln K$ fit is not a derived capacity. The additive constant is refuted as a true constant (Section 5.1), and we ran no existence oracle, so only the shape of the lift is being tested, not its absolute level.
- **Findable $<$ existence; finite- $K \neq$ asymptotic slope.** The solved criterion measures findable capacity, a lower bound on existence capacity; the measured 0.97 is the finite- K slope steepened by the $(\ln \ln \ln K)/\ln 2$ subleading, not the asymptotic 0.721. These two facts pull in opposite, uncontrolled directions and should not be conflated.
- **$\Delta AIC = 3.6$ is positive, not decisive.** As stated in Section 8.5, the double-log form is favored over $\ln K$ but with a relative likelihood of 0.17 on five points; the strong claim is the exclusion of the linear law, not a knockout of $\ln K$.
- **Large- K deceleration may be partly budgetary.** The apparent flattening of $\hat{\alpha}_c$ between $K = 32$ (4.05) and $K = 64$ (4.15) is consistent with the EVT subleading curvature, but a residual component cannot be excluded: the $K = 64$ cell is the most expensive and the slowest to converge, so a small amount of remaining under-budgeting there would depress its crossing slightly. The $K = 8$ timeouts are direct evidence that the high- K cells sit near the budget envelope.

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